

Power Grid parameter estimation using Sparse Identification of Nonlinear Dynamics

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Abstract—The recent discovery of nonlinear system identification via the Sparse Identification of Nonlinear Dynamics (SINDy) method has enjoyed a lot of success across many engineering applications. Due to innovations in sparse regression and compressed sensing, this technique enables tractable identification of both the structure and parameters of a nonlinear dynamical system from data. In the present work, we show the application of SINDy for estimating power-grid parameters. In particular, we demonstrate how SINDy can be used to extract the underlying swing equations from time-series data and thus estimate the critical power-system parameters like rotor inertia and damping coefficients with high degree of accuracy. We demonstrate the results on the Ring-Grid and the IEEE 39-Bus test system.

Index Terms—Power system networks, Sparse identification, data driven methods, Nonlinear dynamical systems.

I. INTRODUCTION

Data-driven identification techniques have a revolutionary impact on the sciences, facilitating a variety of innovations for characterizing high-dimensional data generated from experiments. The success of these techniques is due to the plummeting cost of sensors, data storage and computational resources. Various techniques have been employed for the identification of mathematical models from data, such as, eigen system realization algorithm [23], Kalman filtering methods [26], Dynamic mode decomposition (DMD) [19], [48] and Auto regressive moving averages [8]. These methods provide a principled approach for control design purposes but are not able to capture the underlying dynamics of a given system efficiently [59]. In recent years, neural-network based methods have been proposed [16], [17], [64]. The problems prevalent in these methods is that they lack *interpretability* due to black-box function interpolation and also incur a huge computational cost. Further, these methods depend on massive data-sets for efficient representation of the underlying physics. Methods

employing genetic programming, that use symbolic regression for identification of dynamic model from experimental data have been reported in the recent past, see, e.g., [6], [7], [13], [21], [49]. However, these techniques face various challenges when applied to large-scale systems, sometimes become prohibitively expensive in nature and are prone to overfitting [9].

The basic idea of data-driven model discovery methods is to fit a representation of the given data to a model with fewer terms while simultaneously avoiding overfitting. Symbolic regression based techniques such as SINDy avoid this issue by optimally selecting a parsimonious model and providing a trade-off between model accuracy and complexity via the Pareto analysis [9]. In other words, SINDy selects a set of linear and nonlinear candidate terms that fit a required model to the given data-set from a predefined library. Then the algorithm assigns coefficients to each candidate term, usually zero coefficients for terms that don't contribute to system dynamics and nonzero for the active terms.

The idea of SINDy was derived from [58], where Least Absolute Shrinkage and Selection Operator (LASSO) algorithm was used to promote sparsity using the $\mathcal{L}_1$ -norm for the identification of linear systems. LASSO algorithm was further modified in [9], where the authors made use of a tuning parameter on a linear regression algorithm to construct the right-hand side terms of the governing equations. SINDy algorithm has rapidly gained attention in recent years. Classical SINDy and its variants have been implemented as a system identification tool in biological networks [33], control applications [25], rapid model recovery [42], time-varying aerodynamic systems [29], identification of spatially dependent boundary value problems [53], and also for estimation of partial differential equations [24].

In the context of power-grid networks, there exists sufficient number of dynamical models to represent various components and their interconnections, however, there is an increasing demand for identification of models from data. One of core

problems that a power system engineer faces is the lack of precise knowledge of the power system models. System parameters like generator inertia and damping constants do not always remain constant due to the changing operating conditions and equipment aging. Cascading of power systems with renewable resources and integration with new modern technologies has further added uncertainties to these parameters. Thus, erroneous power system models with inexplicit parameters often lead to an incorrect estimation of system's stability and safety. This can result in system failures and wide-network blackouts as reported in [4], [27]. Normally, Phase Measurement Units (PMU) are used in monitoring of power-grid signals like voltage phase angle and frequencies. While PMU helps in parameter estimation, it has paved the way for data-driven algorithms in power system networks. Identification of models from data in power system finds its roots from the works of [23], [44] and later works like [3], [5], [28], [41], [45], [46], [54], [56], [61], [63]. These algorithms either use steady-state measurements or transient data for system identification. Although these methods yield promising results, they are not efficient in capturing the dynamics of the system for increased load or generation variations [50]. Apart from the above methods, techniques involving system-theoretic measures have been reported for the identification of power-grid models. One such technique is the *Prony* analysis method, where identification of various frequencies and damping coefficients is carried out by performing a decomposition of the given signal into various polynomial equations [1], [14], [15], [20], [22], [60]. Another similar method is the matrix pencil technique in which the Hankel matrix is first constructed from the given data-set followed by the Singular Value Decomposition (SVD) [2], [12], [18]. Subspace identification-based methods have also been proposed [62]. These methods were further revised in [52] employing a linear operator for estimating system dynamics followed by extended subspace methods [51]. A common bottleneck in all these methods is that they are limited in identification of linear system operators only and are highly susceptible to noise. For identifying nonlinear operators from data, several attempts have been made [39], [43], [57]. These methods, however, involve only partial nonlinearity estimation.

Recently, *Koopman* theory has been used for estimation of generator-inertia constants, however, this method is limited to transient event measurements only [55]. One can also find methods based on Artificial Neural-Networks (ANN) [59]. These ANN techniques are used in detection of harmonics [30], [34], [35], faults [10], [11], fault diagnosis in power plants [31], detecting multiple failures in protection systems [37], online identification of multi-machines [59], identifying synchronous generator models [40], and for identifying oscillatory modes in power system [47]. Though these methods are able to handle strong nonlinearities efficiently, they typically involve time-consuming and costly training procedures. In this context, SINDy has proved to be a viable choice, as demonstrated in [28], [46].

Th present work is an effort in a similar direction. We

demonstrate, how SINDy can be used to estimate the inertia and damping constants of power-grid generators. The key contributions of this work can be summarized as under:

- We show how SINDy algorithm can be used to extract the governing equations of power-grid models without any prior knowledge of the underlying nonlinearity (Section III-B).
- We demonstrate how we can use existing knowledge about the system in selecting an appropriate candidate library terms for sparse estimation (Section III-A).

The remaining part of the paper is outlined as follows. In Section II we briefly introduce the general idea of SINDy. In Section III we briefly explain the problem formulation. In Section IV we demonstrate the application of SINDy in power systems models showing the simulation results. In Section V we discuss the conclusions of our work with some future possible modifications.

II. SPARSE IDENTIFICATION OF NONLINEAR DYNAMICS (SINDy)

Consider a continuous-time, nonlinear dynamical system given as;

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t)), \quad (1)$$

where $t \in [0, T]$ with $T \in \mathbb{R}^+$ the final time, $\mathbf{x}(t) \in \mathbb{R}^n$ are the states of the given system and $\mathbf{f} \in (\mathbb{R}^n \rightarrow \mathbb{R}^n)$ denotes the vector-valued nonlinear function. The fundamental assumption of SINDy is that the nonlinear function $\mathbf{f}()$ in (1) can be described by fewer terms in all the space of possible functions, thus promoting sparsity [9]. In the following discussion, we explain the general formulation of SINDy. The procedure begins by collecting the time history of state vector $\mathbf{x}(t)$ in a data-matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$ given as:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1(t_1) & \mathbf{x}_2(t_1) & \dots & \mathbf{x}_n(t_1) \\ \mathbf{x}_1(t_2) & \mathbf{x}_2(t_2) & \dots & \mathbf{x}_n(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{x}_1(t_m) & \mathbf{x}_2(t_m) & \dots & \mathbf{x}_n(t_m) \end{bmatrix}$$

The rows of $\mathbf{X}$ denote different time-instances $t_1, t_2, \dots, t_m$ at which the data is recorded, and the columns denote the number of measurements or the number of sensors used n . Then, the time-derivative of data-matrix $\mathbf{X}$ is obtained, given as:

$$\dot{\mathbf{X}} = \begin{bmatrix} \dot{\mathbf{x}}_1(t_1) & \dot{\mathbf{x}}_2(t_1) & \dots & \dot{\mathbf{x}}_n(t_1) \\ \dot{\mathbf{x}}_1(t_2) & \dot{\mathbf{x}}_2(t_2) & \dots & \dot{\mathbf{x}}_n(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ \dot{\mathbf{x}}_1(t_m) & \dot{\mathbf{x}}_2(t_m) & \dots & \dot{\mathbf{x}}_n(t_m) \end{bmatrix},$$

Note that the amount of data and the type of derivative function plays a significant role in the estimation procedure. Now, a library of possible candidate terms Θ is constructed given as:

$$\Theta(\mathbf{X}) = \begin{bmatrix} | & | & | & | & \dots & | & | & | \\ 1 & \mathbf{X} & \mathbf{X}^{P_2} & \mathbf{X}^{P_3} & \dots & \sin(\mathbf{X}) & \cos(\mathbf{X}) & \dots \\ | & | & | & | & & | & | & | \end{bmatrix},$$

where, $\mathbf{X}^{P_2}, \mathbf{X}^{P_3} \dots$ denote higher order polynomial nonlinear candidates of the given states. The choice of a quadratic polynomial reads:

$$\mathbf{X}^{P_2} = \begin{bmatrix} \mathbf{x}_1^2(t_1) & \mathbf{x}_1(t_1)\mathbf{x}_2(t_1) & \dots & \mathbf{x}_2^2(t_1) & \dots & \mathbf{x}_n^2(t_1) \\ \mathbf{x}_1^2(t_2) & \mathbf{x}_1(t_2)\mathbf{x}_2(t_2) & \dots & \mathbf{x}_2^2(t_2) & \dots & \mathbf{x}_n^2(t_2) \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \mathbf{x}_1^2(t_m) & \mathbf{x}_1(t_m)\mathbf{x}_2(t_m) & \dots & \mathbf{x}_2^2(t_m) & \dots & \mathbf{x}_n^2(t_m) \end{bmatrix}$$

Note that there can be a lot of possible combinations for the number of terms in library function Θ . Since it's assumed that there are only a few active terms that mostly contribute in the right-hand side of (1), the SINDy algorithm estimates these terms by solving the following least-square problem:

$$\dot{\mathbf{X}} = \Theta(\mathbf{X})\Xi, \quad (2)$$

where Ξ solves:

$$\Xi = \arg \min_{\Xi} \|\Theta(\mathbf{X})\Xi - \dot{\mathbf{X}}\|_2^2 + \lambda \|\Xi\|_1. \quad (3)$$

Note that solution matrix Ξ contains the sparse coefficients corresponding to the active nonlinear terms present in the system. The final SINDy model takes the form:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) = \Xi^T (\Theta(\mathbf{x}^T))^T. \quad (4)$$

It should be noted that $\Theta(\mathbf{x}^T)$ is a row vector with each entry as the symbolic function of $\mathbf{x}$ instead of $\Theta(\mathbf{X})$ which is the data matrix. The parameter λ promotes sparsity in the $\mathcal{L}_1$ -norm and it's value is problem-specific which is explained in the next section.

III. FORMULATING SINDY FOR POWER SYSTEMS

In this section, we will demonstrate the application of SINDy algorithm in the context of power-grid networks. Consider a typical power-grid network with n generators, the governing dynamics of the i^{th} generator/oscillator is given as [36]:

$$\mathbf{M}_i \dot{\omega}_i = -\mathbf{D}_i \omega_i - \mathbf{F}_i - \mathbf{f}_i(\delta_i), \quad i \in 1, \dots, n \quad (5)$$

where $\delta_i \in \mathbb{R}^n$ represent the rotor angles, $\mathbf{M}_i, \mathbf{D}_i \in \mathbb{R}^{n \times n}$ are the diagonal matrices containing normalized inertia and damping factors, $\mathbf{f}_i$ is the nonlinear coupling term representing the demanded electrical power of the i -th generator and $\mathbf{F}_i$ is the constant input term representing the input mechanical power.

To estimate the power-grid network parameters via SINDy, we consider two possible cases:

- (Case I:) where the power-system engineer has some prior knowledge about the system.
- (Case II:) where there is no knowledge about the system.

The former involves computation of the unknown parameters i.e., the inertia and damping constants of the generators with some knowledge such as frequency and voltage at corresponding buses via the PMUs, while the later approach is more like a *black-box* method. Both cases are discussed as follows.

A. Case I: SINDy Library with partial knowledge about the system

Several parameters are often known to a power system engineer by observing voltages and rotor angles δ_i using PMU measurements. The system operator also has the knowledge about the generator's mechanical power input and the electrical demands at the buses. As such, we separate-out the known terms on one side and the parameters to be estimated on the other side in (5):

$$\underbrace{\begin{bmatrix} \dot{\omega}_i^{(1)} \\ \vdots \\ \dot{\omega}_i^{(T)} \end{bmatrix}}_{\omega_i} = \underbrace{\begin{bmatrix} -\omega_i^{(1)} & -c_i^{(1)} \\ \vdots & \vdots \\ -\omega_i^{(T)} & -c_i^{(T)} \end{bmatrix}}_{\Theta_i} \underbrace{\begin{bmatrix} \mathbf{D}_i \\ \mathbf{M}_i \\ \frac{1}{\mathbf{M}_i} \end{bmatrix}}_{\Xi_i} \quad (6)$$

where

$$c_i^\tau = \mathbf{F}_i + \mathbf{f}_i(\delta_i^\tau), \quad \tau \in 1, 2, \dots, T. \quad (7)$$

Equation (6) represent the relationship between the generator inertia, damping coefficients and the observed PMU data. Note that (6) represents the discrete version of (5) where the rows represent different time instances. The entries of matrix Θ_i can be obtained from the PMU values and the corresponding left side entries of (6) can be obtained by taking the derivative using either finite difference method for noiseless data or polynomial interpolation method for noisy data [9]. The next step is to compute the unknown parameters Ξ_i using the SINDy procedure discussed above by solving the following optimization problem which also corresponds to the LASSO algorithm [58]:

$$\Xi_i = \arg \min_{\Xi_i} \|\Theta_i(\delta, \omega)\Xi_i - \dot{\omega}_i\|_2^2 + \lambda \|\Xi_i\|_1, \quad (8)$$

A low value of λ results in less sparse solution leading to an incorrect parameter estimation as there can be multiple solutions that lead to the similar dynamics. On the other hand, a high value of λ favors an overly sparse solution with most of the parameters estimated to be zero. As such, the choice of λ plays a significant role in the estimation process.

B. Case II: SINDy library with no knowledge about the system

In this case, we assume that the user has no knowledge of the governing equations corresponding to the entries in Θ_i . For this case, the candidate libraries can constructed intelligently by using some domain knowledge into practice. For instance, the constant terms can be first selected to correspond to the mechanical input terms in (5). Then, linear candidates in δ and ω can be selected along-with nonlinear candidate functions such as polynomial terms and the trigonometric nonlinearities such as *sin* and *cosine* terms. Note that a good choice of candidate nonlinearities can facilitate an accurate approximation of the underlying dynamics. The procedure discussed herein can be applied to a general power-grid network where the candidate library can be tailored for the specific network architecture.

The final SINDy algorithm for power systems is now summarized as under:

Algorithm 1 : SINDy for Power-Grid Networks [9]

Input: Rotor-angle data δ_i and frequency data ω_i

Output: Sparse coefficient matrix Ξ_i

- 1: Compute the numerical derivative: $\dot{\delta}_i$ and $\dot{\omega}_i$
 - 2: Construct the library Θ_i (as discussed above)
 - 3: Set tuning parameter λ
 - 4: Solve (8) to obtain Ξ_i .
-

IV. SIMULATION RESULTS

In this section, we describe the application of SINDy on two power-grid models corresponding to the two case scenarios discussed in section (III).

A. Ring-Grid model

Ring-Grid model represents a ring type arrangement of n identical generators connected to a common bus. The model and the parameters are taken from [32]. The governing equation is of the form (5) where $\mathbf{f}_i(\delta)$ given by:

$$\mathbf{f}_i(\delta) = b_1 \sin(\delta_i) + b_2 [\sin(\delta_i - \delta_{i+1}) + \sin(\delta_i) - \delta_{i-1}], \quad (9)$$

where, b_1 and b_2 are constant entries. For the Ring-Grid model, we assumed that we don't have any knowledge of the governing equations but only access to the simulation data. We collected the data by integrating the nonlinear ODE (5) via the MATLAB's built-in *ode45* solver with a fixed time step of 0.01. Then, we constructed the numerical derivatives via finite difference method. The matrix Θ_i was precomputed as under:

$$\Theta_i(\delta, \omega) = \begin{bmatrix} \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \delta_i & \omega_i & \delta_i^2 & \omega_i^2 & \sin(\delta_i) & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}. \quad (10)$$

With the data and candidate libraries build, we used the SINDy algorithm (1) with $\lambda = 0.1$ to obtain the sparse coefficient matrix Ξ . The actual differential equation is given as:

$$\ddot{\delta}_i + 0.5\dot{\delta}_i = -0.52 - 3\sin(\delta_i) - 100[\sin(\delta_i - \delta_{i+1}) + \sin(\delta_i - \delta_{i-1})]. \quad (11)$$

whereas, the identified equation from SINDy is given as:

$$\begin{aligned} \ddot{\delta}_i + 0.5\dot{\delta}_i = & -3\sin(\delta_i) - 100[\sin(\delta_i - \delta_{i+1}) + \sin(\delta_i - \delta_{i-1})] \\ & - 0.53 + 0(\delta) + 0(\delta^2) + 0(\delta\dot{\delta}) + 0(\dot{\delta}^2) + 0(\delta^3) \\ & + 0(\dot{\delta}^3) + 0(\delta^2\dot{\delta}) + 0(\delta\dot{\delta}\dot{\delta}) + 0(\dot{\delta}\dot{\delta}^2) + 0(\sin(\dot{\delta})). \end{aligned} \quad (12)$$

As can be observed that SINDy not only identified the correct terms of the underlying dynamics but also ruled out the other nonlinear terms such as quadratic or cubic by assigning zero coefficients to them. The reconstructed output response is shown in Fig. 1. We also tested SINDy by taking different values of the right-hand side in (9), where the algorithm was able to recover the coefficients with higher accuracy as enlisted in Table (I).

TABLE I
NUMERICAL COMPARISON OF TRUE AND IDENTIFIED PARAMETERS FOR RING GRID MODEL

Parameters	Case I		Case II		Case III	
	True	Identified	True	Identified	True	Identified
D	0.25	0.25	0.5	0.5	0.95	0.94
b_1	2	2	3	3	0.45	0.45
P_m	0.02	0.019	0.52	0.53	0.8	0.79

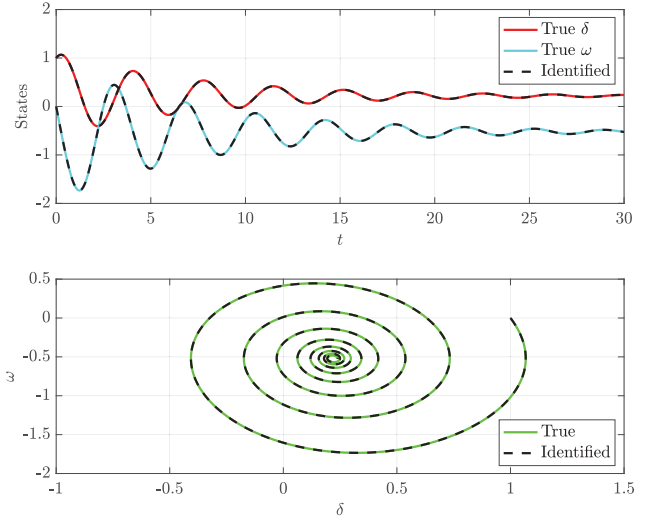


Fig. 1. Comparison of true and identified dynamics of Ring-Grid model: (above:) output response; (below:) phase portrait

B. IEEE 39 bus model

To demonstrate SINDy algorithm on a more practical power system model, we considered the standard IEEE 39-Bus model [38]. The data was generated using MATPOWER library [65] and pg-sync-models [38] wherein an effective-network (EN) model representation was obtained. The time-snapshots were again taken using *ode45*.

In this case, we assumed that the underlying nonlinear terms (cf. 7) were known and we constructed the library (10) accordingly. The first three column of library were the same as given in (10) and the next candidate in the library was constructed according to the terms in the original governing equation that are available via PMU reading and given as:

$$\mathbf{f}_i(\delta) = - \sum_{j=i; j \neq i}^n \mathbf{K}_{ij} \sin(\delta_i - \delta_j - \gamma_{ij}). \quad (13)$$

Here the constant terms $\mathbf{K}_{ij}$ and γ_{ij} depend upon the solution of power-flow equation. For this case, SINDy identified all the inertia and damping constants correctly as shown in Fig. 3. Furthermore, a perfect reconstruction of the swing dynamics was obtained for all generators as depicted in Fig. 2. For the IEEE case, the value of the hyper-parameter λ was kept the same.

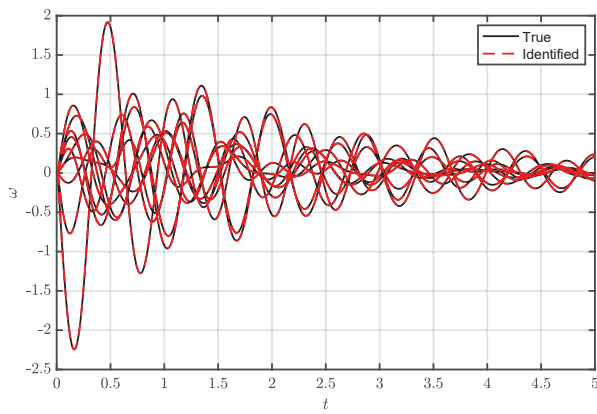


Fig. 2. Comparison of output response dynamics for IEEE 39-Bus model

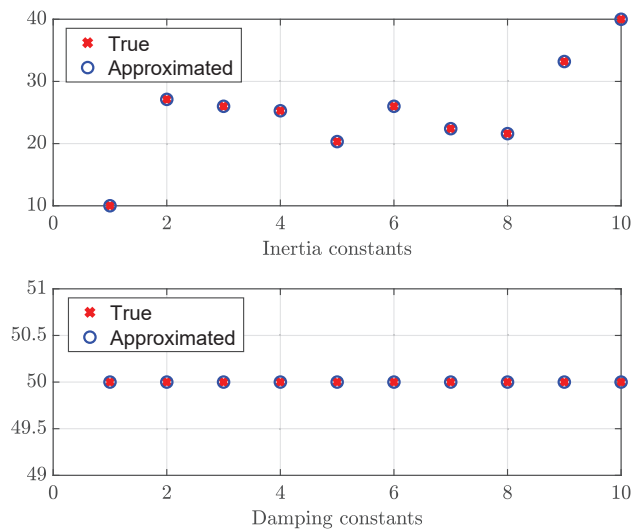


Fig. 3. Comparison of true and identified parameters of IEEE 39-Bus model

V. CONCLUSION AND FUTURE SCOPE

In this paper, we have used SINDy as a data-driven technique to identify the system dynamics and system parameters of various power-grid models. The simulation results depict that the SINDy algorithm estimates the unknown parameters with higher accuracy. The computational requirements are also less than the traditional machine-learning approaches. However, SINDy requires an efficient coordinate system for proper scaling of candidate library matrix. The library grows exponentially for higher-dimensional coordinate system. Therefore, in such scenarios data-compression techniques such as an auto-encoder can be used for the identification of a low-dimensional coordinate system for power-grid models.

REFERENCES

- [1] Anas Almunif and Lingling Fan. Pmu measurements for oscillation monitoring: Connecting prony analysis with observability. In *2019 IEEE Power & Energy Society General Meeting (PESGM)*, pages 1–5. IEEE, 2019.
- [2] Anas Almunif, Lingling Fan, and Zhixin Miao. A tutorial on data-driven eigenvalue identification: Prony analysis, matrix pencil, and eigensystem realization algorithm. *International Transactions on Electrical Energy Systems*, 30(4):e12283, 2020.
- [3] Krishna K Anaparthi, Balarko Chaudhuri, Nina F Thornhill, and Bikash C Pal. Coherency identification in power systems through principal component analysis. *IEEE transactions on power systems*, 20(3):1658–1660, 2005.
- [4] Göran Andersson et al. Causes of the 2003 major grid blackouts in north america and europe, and recommended means to improve system dynamic performance. *IEEE transactions on Power Systems*, 20(4):1922–1928, 2005.
- [5] MA Mohd Ariff and Bikash C Pal. Coherency identification in inter-connected power system—an independent component analysis approach. *IEEE Transactions on Power Systems*, 28(2):1747–1755, 2012.
- [6] Douglas Adriano Augusto and Helio JC Barbosa. Symbolic regression via genetic programming. In *Proceedings. Vol. 1. Sixth Brazilian Symposium on Neural Networks*, pages 173–178. IEEE, 2000.
- [7] Josh Bongard and Hod Lipson. Automated reverse engineering of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 104(24):9943–9948, 2007.
- [8] Peter J Brockwell and Richard A Davis. *Introduction to time series and forecasting*. Springer, 2002.
- [9] Steven L Brunton, Joshua L Proctor, and J Nathan Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the national academy of sciences*, 113(15):3932–3937, 2016.
- [10] Ghendy Cardoso, Jacqueline GisÈle Rolim, and Hans Helmut Zurn. Identifying the primary fault section after contingencies in bulk power systems. *IEEE Transactions on Power Delivery*, 23(3):1335–1342, 2008.
- [11] Ghendy Cardoso, Jacqueline GisÈle Rolim, and HH Zurn. Application of neural-network modules to electric power system fault section estimation. *IEEE Transactions on Power Delivery*, 19(3):1034–1041, 2004.
- [12] Mariesa L Crow and A Singh. The matrix pencil for power system modal extraction. *IEEE Transactions on Power Systems*, 20(1):501–502, 2005.
- [13] John Duffy and Jim Engle-Warnick. Using symbolic regression to infer strategies from experimental data. In *Evolutionary computation in Economics and Finance*, pages 61–82. Springer, 2002.
- [14] Lingling Fan. Data fusion-based distributed prony analysis. *Electric Power Systems Research*, 143:634–642, 2017.
- [15] Lingling Fan and Yasser Wehbe. Extended kalman filtering based real-time dynamic state and parameter estimation using pmu data. *Electric Power Systems Research*, 103:168–177, 2013.
- [16] Keisuke Fujii and Yoshinobu Kawahara. Supervised dynamic mode decomposition via multitask learning. *Pattern Recognition Letters*, 122:7–13, 2019.
- [17] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep learning*. MIT press, 2016.
- [18] Lisa L Grant and Mariesa L Crow. Comparison of matrix pencil and prony methods for power system modal analysis of noisy signals. In *2011 North American power symposium*, pages 1–7. IEEE, 2011.
- [19] Jonathan H. Tu, Clarence W. Rowley, Dirk M. Luchtenburg, Steven L. Brunton, and J. Nathan Kutz. On dynamic mode decomposition: Theory and applications. *Journal of Computational Dynamics*, 1(2):391–421, 2014.
- [20] John F Hauer, CJ Demeure, and LL Scharf. Initial results in prony analysis of power system response signals. *IEEE Transactions on power systems*, 5(1):80–89, 1990.
- [21] Nguyen Xuan Hoai. Solving the symbolic regression problem with tree-adjunct grammar guided genetic programming: The preliminary results. In *In Proceedings of Congress on Evolutionary Computation (CEC’2002)*, Hawaii. Citeseer, 2002.
- [22] Md Ismail Hossain, Md Shafiullah, Mohammad Abido, and Md Al Emran. Online monitoring of inter-area oscillations and damping of power systems employing prony analysis. In *2018 10th international conference on electrical and computer engineering (ICECE)*, pages 269–272. IEEE, 2018.
- [23] Jer-Nan Juang and Richard S Pappa. An eigensystem realization algorithm for modal parameter identification and model reduction. *Journal of guidance, control, and dynamics*, 8(5):620–627, 1985.
- [24] Kadierdan Kaheman, J Nathan Kutz, and Steven L Brunton. Sindy-pi: a robust algorithm for parallel implicit sparse identification of nonlinear dynamics. *Proceedings of the Royal Society A*, 476(2242):20200279, 2020.

- [25] Eurika Kaiser, J Nathan Kutz, and Steven L Brunton. Sparse identification of nonlinear dynamics for model predictive control in the low-data limit. *Proceedings of the Royal Society A*, 474(2219):20180335, 2018.
- [26] Rudolph Emil Kalman. A new approach to linear filtering and prediction problems. 1960.
- [27] Dmitry N Kosterev, Carson W Taylor, and William A Mittelstadt. Model validation for the august 10, 1996 wscs system outage. *IEEE transactions on power systems*, 14(3):967–979, 1999.
- [28] Subhash Lakshminarayana, Saurav Sthapit, and Carsten Maple. Application of physics-informed machine learning techniques for power grid parameter estimation. *Sustainability*, 14(4):2051, 2022.
- [29] Shanwu Li, Eurika Kaiser, Shujin Laima, Hui Li, Steven L Brunton, and J Nathan Kutz. Discovering time-varying aerodynamics of a prototype bridge by sparse identification of nonlinear dynamical systems. *Physical Review E*, 100(2):022220, 2019.
- [30] Hsiung Cheng Lin. Intelligent neural network-based fast power system harmonic detection. *IEEE Transactions on Industrial Electronics*, 54(1):43–52, 2007.
- [31] Liangyu Ma, Yongguang Ma, and Kwang Y Lee. An intelligent power plant fault diagnostics for varying degree of severity and loading conditions. *IEEE Transactions on Energy Conversion*, 25(2):546–554, 2010.
- [32] Muhammad Haris Malik, Domenico Borzacchiello, Francisco Chinesta, and Pedro Diez. Reduced order modeling for transient simulation of power systems using trajectory piece-wise linear approximation. *Advanced Modeling and Simulation in Engineering Sciences*, 3(1):1–18, 2016.
- [33] Niall M Mangan, Steven L Brunton, Joshua L Proctor, and J Nathan Kutz. Inferring biological networks by sparse identification of nonlinear dynamics. *IEEE Transactions on Molecular, Biological and Multi-Scale Communications*, 2(1):52–63, 2016.
- [34] Joy Mazumdar, Ronald G Harley, Frank C Lambert, and Ganesh K Venayagamoorthy. Neural network based method for predicting nonlinear load harmonics. *IEEE Transactions on Power Electronics*, 22(3):1036–1045, 2007.
- [35] Joy Mazumdar, Ronald G Harley, Frank C Lambert, Ganesh Kumar Venayagamoorthy, and Marty L Page. Intelligent tool for determining the true harmonic contribution of a customer in a power distribution network. *IEEE Transactions on Industry Applications*, 44(5):1477–1485, 2008.
- [36] Federico Milano. *Power system modelling and scripting*. Springer Science & Business Media, 2010.
- [37] Michael Negnevitsky and Vsevolod Pavlovsky. Neural networks approach to online identification of multiple failures of protection systems. *IEEE Transactions on Power Delivery*, 20(2):588–594, 2005.
- [38] Takashi Nishikawa and Adilson E Motter. Comparative analysis of existing models for power-grid synchronization. *New Journal of Physics*, 17(1):015012, 2015.
- [39] Harish J Palanthandalam-Madapusi, Seth Lacy, Jesse B Hoagg, and Dennis S Bernstein. Subspace-based identification for linear and nonlinear systems. In *Proceedings of the 2005, American Control Conference, 2005.*, pages 2320–2334. IEEE, 2005.
- [40] Jung-Wook Park, Ganesh K Venayagamoorthy, and Ronald G Harley. Mlp/rbf neural-networks-based online global model identification of synchronous generator. *IEEE Transactions on Industrial Electronics*, 52(6):1685–1695, 2005.
- [41] Robin Podmore. Identification of coherent generators for dynamic equivalents. *IEEE Transactions on Power Apparatus and Systems*, (4):1344–1354, 1978.
- [42] Markus Quade, Markus Abel, J Nathan Kutz, and Steven L Brunton. Sparse identification of nonlinear dynamics for rapid model recovery. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 28(6):063116, 2018.
- [43] Thordur Runolfsson. Output dynamic mode decomposition: An extension of dynamic mode decomposition based on output functional expansions. In *2018 IEEE Conference on Decision and Control (CDC)*, pages 7148–7152. IEEE, 2018.
- [44] J.J. Sanchez-Gasca, C.J. Bridenbaugh, C.E.J. Bowler, and J.S. Edmonds. Trajectory sensitivity based identification of synchronous generator and excitation system parameters. *IEEE Transactions on Power Systems*, 3(4):1814–1822, 1988.
- [45] Juan Sanchez-Gasca, Daniel Trudnowski, Emilio Barocio, John Hauer, John W Pierre, Claudio Canizares, Henry Huang, Tuomas Rauhala, Joe H Chow, Innocent Kamwa, et al. Identification of electromechanical modes in power systems. In *IEEE Task Force Report, Special Publication TP462*. IEEE, 2012.
- [46] Andrija T Sarić, Aleksandar A Sarić, Mark K Transtrum, and Aleksandar M Stanković. Symbolic regression for data-driven dynamic model refinement in power systems. *IEEE Transactions on Power Systems*, 36(3):2390–2402, 2020.
- [47] Rahul Satheesh, Nashida Chakkungal, Sunitha Rajan, Manu Madhavan, and Hassan Haes Alhelou. Identification of oscillatory modes in power system using deep learning approach. *IEEE Access*, 10:16556–16565, 2022.
- [48] Peter J Schmid. Dynamic mode decomposition of numerical and experimental data. *Journal of fluid mechanics*, 656:5–28, 2010.
- [49] Michael Schmidt and Hod Lipson. Distilling free-form natural laws from experimental data. *science*, 324(5923):81–85, 2009.
- [50] Fred C. Schweppe and Douglas B. Rom. Power system static-state estimation, part ii: Approximate model. *IEEE Transactions on Power Apparatus and Systems*, PAS-89(1):125–130, 1970.
- [51] Pranav Sharma, Venkataramana Ajjarapu, and Umesh Vaidya. Data-driven identification of nonlinear power system dynamics using output-only measurements. *IEEE Transactions on Power Systems*, 2021.
- [52] Pranav Sharma, Bowen Huang, Venkataramana Ajjarapu, and Umesh Vaidya. Data-driven identification and prediction of power system dynamics using linear operators. In *2019 IEEE Power & Energy Society General Meeting (PESGM)*, pages 1–5. IEEE, 2019.
- [53] Daniel E Shea, Steven L Brunton, and J Nathan Kutz. Sindy-bvp: Sparse identification of nonlinear dynamics for boundary value problems. *Physical Review Research*, 3(2):023255, 2021.
- [54] Alex M Stanković, Aleksandar A Sarić, Andrija T Sarić, and Mark K Transtrum. Data-driven symbolic regression for identification of nonlinear dynamics in power systems. In *2020 IEEE Power & Energy Society General Meeting (PESGM)*, pages 1–5. IEEE, 2020.
- [55] Yoshihiko Susuki, Ryo Hamasaki, and Atsushi Ishigame. Estimation of power system inertia using nonlinear koopman modes. In *2018 IEEE Power & Energy Society General Meeting (PESGM)*, pages 1–5. IEEE, 2018.
- [56] Yoshihiko Susuki and Igor Mezic. Nonlinear koopman modes and coherency identification of coupled swing dynamics. *IEEE Transactions on Power Systems*, 26(4):1894–1904, 2011.
- [57] Naoya Takeishi, Yoshinobu Kawahara, and Takehisa Yairi. Subspace dynamic mode decomposition for stochastic koopman analysis. *Physical Review E*, 96(3):033310, 2017.
- [58] Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological)*, 58(1):267–288, 1996.
- [59] Sheela Tiwari, Ram Naresh, and Rameshwar Jha. Comparative study of backpropagation algorithms in neural network based identification of power system. *AIRCC's International Journal of Computer Science and Information Technology*, 5(4):93–107, 2013.
- [60] Daniel J Trudnowski, JM Johnson, and John F Hauer. Making prony analysis more accurate using multiple signals. *IEEE Transactions on power systems*, 14(1):226–231, 1999.
- [61] Ilya Tyuryukanov, Marjan Popov, Mart AMM van der Meijden, and Vladimir Terzija. Slow coherency identification and power system dynamic model reduction by using orthogonal structure of electromechanical eigenvectors. *IEEE Transactions on Power Systems*, 36(2):1482–1492, 2020.
- [62] Peter Van Overschee and BL De Moor. *Subspace identification for linear systems: Theory—Implementation—Applications*. Springer Science & Business Media, 2012.
- [63] Dechang Yang, Christian Rehtanz, Yong Li, and Deyu Cai. Identification of dominant oscillation mode using complex singular value decomposition method. *Electric power systems research*, 83(1):227–236, 2012.
- [64] Enoch Yeung, Soumya Kundu, and Nathan Hodas. Learning deep neural network representations for koopman operators of nonlinear dynamical systems. In *2019 American Control Conference (ACC)*, pages 4832–4839. IEEE, 2019.
- [65] Ray Daniel Zimmerman, Carlos Edmundo Murillo-Sánchez, and Robert John Thomas. Matpower: Steady-state operations, planning, and analysis tools for power systems research and education. *IEEE Transactions on power systems*, 26(1):12–19, 2010.